Least squares and projection

Let A be an $m \times n$ real matrix and $\mathbf{b} \in \mathbb{R}^m$.

Least-squares solution: $\hat{\mathbf{x}} \in \mathbb{R}^n$ minimizes $\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^m ((A\mathbf{x})_i - b_i)^2$ over $\mathbf{x} \in \mathbb{R}^n$. Minimizing the norm gives the same solutions.

Fitted vector and residual: $\hat{\mathbf{b}} = A\hat{\mathbf{x}}$ and $\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}} = \mathbf{b} - A\hat{\mathbf{x}}$.

Projection: $\hat{\mathbf{b}} = \text{proj}_{\text{col}(A)}(\mathbf{b})$ is the unique closest vector in $\text{col}(A)$; $\mathbf{r} \perp \text{col}(A)$. A least-squares solution always exists. If $\mathbf{b} \in \text{col}(A)$, the minimum error is zero and the solutions solve $A\mathbf{x} = \mathbf{b}$.

Normal equations and uniqueness

Column orthogonality: $A^T\mathbf{r} = \mathbf{0}$ exactly when $\mathbf{r}$ is orthogonal to every column of A; $\text{col}(A)^\perp = \text{null}(A^T)$.

Normal equations: $\hat{\mathbf{x}}$ is a least-squares solution if and only if $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$, equivalently

$$A^T A\hat{\mathbf{x}} = A^T\mathbf{b}.$$

Uniqueness: The fitted vector is unique, but coefficients need not be: if $\mathbf{z} \in \text{null}(A)$, then $A(\hat{\mathbf{x}} + \mathbf{z}) = A\hat{\mathbf{x}}$. $A^T A$ is invertible if and only if the columns of A are linearly independent. In that case the least-squares solution is unique and

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T\mathbf{b}.$$

Linear regression notation

For data (t_i, y_i), $i = 1, \ldots, m$, the line model is $\hat{y}_i = c_0 + c_1 t_i$, where c_0 is the **intercept** and c_1 the **slope**. The **residual** is $r_i = y_i - \hat{y}_i$.

Design matrix: Each column of X represents a feature. For the line model,

$$X = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}.$$

The coefficient vector is $\mathbf{c} = \begin{bmatrix} c_0 & c_1 \end{bmatrix}^T$.

The response vector is $\mathbf{y} = \begin{bmatrix} y_1 & \cdots & y_m \end{bmatrix}^T$. The fitted values are $X\hat{\mathbf{c}}$, where $\hat{\mathbf{c}}$ minimizes $\|X\mathbf{c} - \mathbf{y}\|^2 = \sum_{i=1}^m r_i^2$.